

Questions with Answer Keys

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Q1 - 2024 (01 Feb Shift 1)

Let 3, a , b , c be in A.P. and 3 , $a - 1$, $b + 1$, $c + 9$ be in G.P. Then, the arithmetic mean of a , b and c is :

(1) -4

(2) -1

(3) 13

(4) 11

Q2 - 2024 (01 Feb Shift 1)

Let 3, 7, 11, 15, ..., 403 and 2, 5, 8, 11, ..., 404 be two arithmetic progressions. Then the sum, of the common terms in them, is equal to

Q3 - 2024 (01 Feb Shift 2)

Let S_n denote the sum of the first n terms of an arithmetic progression. If $S_{10} = 390$ and the ratio of the tenth and the fifth terms is 15 : 7, then $S_{15} - S_5$ is equal to:

(1) 800

(2) 890

(3) 790

(4) 690

Q4 - 2024 (01 Feb Shift 2)

If three successive terms of a G.P. with common ratio r ($r > 1$) are the lengths of the sides of a triangle and $[r]$ denotes the greatest integer less than or equal to r , then $3[r] + [-r]$ is equal to :

Q5 - 2024 (27 Jan Shift 1)

The number of common terms in the progressions 4, 9, 14, 19, ..., up to 25th term and 3, 6, 9, 12, up to 37th term is :

(1) 9

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- (2) 5
- (3) 7
- (4) 8
- (1) 9
- (2) 5
- (3) 7
- (4) 8

Q6 - 2024 (27 Jan Shift 1)

If

$$8 = 3 + \frac{1}{4}(3 + p) + \frac{1}{4^2}(3 + 2p) + \frac{1}{4^3}(3 + 3p) + \dots \infty,$$

then the value of p is _____

Q7 - 2024 (27 Jan Shift 2)

The 20th term from the end of the progression $20, 19\frac{1}{4}, 18\frac{1}{2}, 17\frac{3}{4}, \dots, -129\frac{1}{4}$ is :-

- (1) -118
- (2) -110
- (3) -115
- (4) -100

Q8 - 2024 (29 Jan Shift 1)

If in a G.P. of 64 terms, the sum of all the terms is 7 times the sum of the odd terms of the G.P, then the common ratio of the G.P. is equal to

- (1) 7
- (2) 4
- (3) 5
- (4) 6

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Q9 - 2024 (29 Jan Shift 1)

In an A.P., the sixth terms $a_6 = 2$. If the $a_1 a_4 a_5$ is the greatest, then the common difference of the A.P., is equal to

(1) $\frac{3}{2}$

(2) $\frac{8}{5}$

(3) $\frac{2}{3}$

(4) $\frac{5}{8}$

Q10 - 2024 (29 Jan Shift 2)

If $\log_e a, \log_e b, \log_e c$ are in an A.P. and $\log_e a - \log_e 2b, \log_e 2b - \log_e 3c, \log_e 3c - \log_e a$ are also in an A.P, then $a : b : c$ is equal to

(1) $9 : 6 : 4$

(2) $16 : 4 : 1$

(3) $25 : 10 : 4$

(4) $6 : 3 : 2$

Q11 - 2024 (29 Jan Shift 2)

If each term of a geometric progression $a_1, a_2, a_3, \dots$ with $a_1 = \frac{1}{8}$ and $a_2 \neq a_1$, is the arithmetic mean of the next two terms and $S_n = a_1 + a_2 + \dots + a_n$, then $S_{20} - S_{18}$ is equal to

(1) 2^{15}

(2) -2^{18}

(3) 2^{18}

(4) -2^{15}

Q12 - 2024 (30 Jan Shift 1)

Let S_n denote the sum of first n terms an arithmetic progression. If $S_{20} = 790$ and $S_{10} = 145$, then $S_{15} - S_5$ is :

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(1) 395

(2) 390

(3) 405

(4) 410

Q13 - 2024 (30 Jan Shift 1)

Let $\alpha = 1^2 + 4^2 + 8^2 + 13^2 + 19^2 + 26^2 + \dots$ upto 10 terms and $\beta = \sum_{n=1}^{10} n^4$. If $4\alpha - \beta = 55k + 40$, then k is equal to

Q14 - 2024 (30 Jan Shift 2)

Let a and b be two distinct positive real numbers. Let 11th term of a GP, whose first term is a and third term is b , is equal to p^{th} term of another GP, whose first term is a and fifth term is b . Then p is equal to

(1) 20

(2) 25

(3) 21

(4) 24

Q15 - 2024 (30 Jan Shift 2)

Let S_n be the sum to n -terms of an arithmetic progression 3, 7, 11, $\dots$

If $40 < \left(\frac{6}{n(n+1)} \sum_{k=1}^n s_k \right) < 42$, then n equals _____.

Q16 - 2024 (31 Jan Shift 1)

The sum of the series $\frac{1}{1-3 \cdot 1^2+1^4} + \frac{2}{1-3 \cdot 2^2+2^4} + \frac{3}{1-3 \cdot 3^2+3^4} + \dots$ up to 10 terms is

(1) $\frac{45}{109}$ (2) $-\frac{45}{109}$ (3) $\frac{55}{109}$ (4) $-\frac{55}{109}$

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Questions with Answer Keys

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Q17 - 2024 (31 Jan Shift 2)

Let 2^{nd} , 8^{th} and 44^{th} terms of a non-constant A.P. be respectively the 1^{st} , 2^{nd} and 3^{rd} terms of G.P. If the first term of A.P. is 1 then the sum of first 20 terms is equal to-

(1) 980

(2) 960

(3) 990

(4) 970

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Answer Key

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Solutions

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Q1

$$3, a, b, c \rightarrow \text{A.P} \Rightarrow 3, 3 + d, 3 + 2d, 3 + 3d$$

$$3, a - 1, b + 1, c + 9 \rightarrow \text{G.P} \Rightarrow 3, 2 + d, 4 + 2d, 12 + 3d$$

$$a = 3 + d \quad (2 + d)^2 = 3(4 + 2d)$$

$$b = 3 + 2d \quad d = 4, -2$$

$$c = 3 + 3d$$

$$\text{If } d = 4 \quad \text{G.P} \Rightarrow 3, 6, 12, 24$$

$$a = 7$$

$$b = 11$$

$$c = 15$$

$$\frac{a + b + c}{3} = 11$$

Q2

$$3, 7, 11, 15, \dots, 403$$

$$2, 5, 8, 11, \dots, 404$$

$$\text{LCM}(4, 3) = 12$$

$$11, 23, 35, \dots \text{ let } (403)$$

$$403 = 11 + (n - 1) \times 12$$

$$\frac{392}{12} = n - 1$$

$$33 \cdot 66 = n$$

$$n = 33$$

$$\text{Sum } \frac{33}{2} (22 + 32 \times 12)$$

$$= 6699$$

Q3

$$S_{10} = 390$$

$$\frac{10}{2} [2a + (10 - 1)d] = 390$$

$$\Rightarrow 2a + 9d = 78 \dots (1)$$

$$\frac{t_{10}}{t_5} = \frac{15}{7} \Rightarrow \frac{a+9d}{a+4d} = \frac{15}{7} \Rightarrow 8a = 3d \dots (2)$$

$$\text{From (1) \& (2), } a = 3 \text{ \& } d = 8$$

$$S_{15} - S_5 = \frac{15}{2} (6 + 14 \times 8) - \frac{5}{2} (6 + 4 \times 8)$$

$$= \frac{15 \times 118 - 5 \times 38}{2} = 790$$

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Solutions

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Q4

 $a, ar, ar^2 \rightarrow \text{G.P.}$ Sum of any two sides $>$ third side

$$a + ar > ar^2, a + ar^2 > ar, ar + ar^2 > a$$

$$r^2 - r - 1 < 0$$

$$r \in \left(\frac{1 - \sqrt{5}}{2}, \frac{1 + \sqrt{5}}{2} \right) \dots (1)$$

$$r^2 - r + 1 > 0$$

always true

$$r^2 + r - 1 > 0$$

$$r \in \left(-\infty, -\frac{1 + \sqrt{5}}{2} \right) \cup \left(\frac{-1 + \sqrt{5}}{2}, \infty \right) \dots (2)$$

Taking intersection of (1), (2)

$$r \in \left(-\frac{1 + \sqrt{5}}{2}, \frac{1 + \sqrt{5}}{2} \right)$$

As $r > 1$

$$r \in \left(1, \frac{1 + \sqrt{5}}{2} \right)$$

$$[r] = 1, [-r] = -2$$

$$3[r] + [-r] = 1$$

Q5

4, 9, 14, 19, ..., up to 25th term

$$T_{25} = 4 + (25 - 1)5 = 4 + 120 = 124$$

3, 6, 9, 12, ..., up to 37th term

$$T_{37} = 3 + (37 - 1)3 = 3 + 108 = 111$$

Common difference of Ist series $d_1 = 5$ Common difference of IInd series $d_2 = 3$

First common term = 9, and

their common difference = 15 (LCM of d_1 and d_2)

then common terms are

9, 24, 39, 54, 69, 84, 99

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Solutions

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Q6

$$8 = \frac{3}{1-\frac{1}{4}} + \frac{p \cdot \frac{1}{4}}{\left(1-\frac{1}{4}\right)^2}$$

(sum of infinite terms of A.G.P = $\frac{a}{1-r} + \frac{dr}{(1-r)^2}$)

$$\Rightarrow \frac{4p}{9} = 4 \Rightarrow p = 9$$

Q7

$$20, 19\frac{1}{4}, 18\frac{1}{2}, 17\frac{3}{4}, \dots, -129\frac{1}{4}$$

This is A.P. with common difference

$$d_1 = -1 + \frac{1}{4} = -\frac{3}{4}$$

$$-129\frac{1}{4}, \dots, 19\frac{1}{4}, 20$$

This is also A.P. $a = -129\frac{1}{4}$ and $d = \frac{3}{4}$

Required term =

$$-129\frac{1}{4} + (20 - 1) \left(\frac{3}{4} \right)$$

$$= -129 - \frac{1}{4} + 15 - \frac{3}{4} = -115$$

Q8

$$a + ar + ar^2 + ar^3 + \dots + ar^{63}$$

$$= 7(a + ar^2 + ar^4 + \dots + ar^{62})$$

$$\frac{a(1 - r^{64})}{1 - r} = \frac{7a(1 - r^{64})}{1 - r^2}$$

$$\Rightarrow \frac{1 - r}{1 - r^2} = \frac{1}{r}$$

$$r = 6$$

Q9

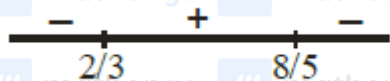
$$a_6 = 2 \Rightarrow a + 5d = 2$$

$$a_1 a_4 a_5 = a(a + 3d)(a + 4d)$$

$$= (2 - 5d)(2 - 2d)(2 - d)$$

$$f(d) = 8 - 32d + 34d^2 - 20d + 30d^2 - 10d^3$$

$$f'(d) = -2(5d - 8)(3d - 2)$$



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Solutions

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$$d = \frac{8}{5}$$

Q10

$\log_e a, \log_e b, \log_e c$ are in A.P.

$$\therefore b^2 = ac \dots (i)$$

Also

$\log_e \left(\frac{a}{2b} \right), \log_e \left(\frac{2b}{3c} \right), \log_e \left(\frac{3c}{a} \right)$ are in A.P.

$$\left(\frac{2b}{3c} \right)^2 = \frac{a}{2b} \times \frac{3c}{a}$$

$$\frac{b}{c} = \frac{3}{2}$$

Putting in eq. (i) $b^2 = a \times \frac{2b}{3}$

$$\frac{a}{b} = \frac{3}{2}$$

$$a : b : c = 9 : 6 : 4$$

Q11

Let r' th term of the GP be ar^{n-1} . Given,

$$2a_r = a_{r+1} + a_{r+2}$$

$$2ar^{n-1} = ar^n + ar^{n+1}$$

$$\frac{2}{r} = 1 + r$$

$$r^2 + r - 2 = 0$$

Hence, we get, $r = -2$ (as $r \neq 1$)

So, $S_{20} - S_{18} = (\text{Sum upto 20 terms}) - (\text{Sum upto}$

18 terms) = $T_{19} + T_{20}$

$$T_{19} + T_{20} = ar^{18}(1 + r)$$

Putting the values $a = \frac{1}{8}$ and $r = -2$;

we get $T_{19} + T_{20} = -2^{15}$

Q12

$$S_{20} = \frac{20}{2} [2a + 19d] = 790$$

$$2a + 19d = 79 \dots (i)$$

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Solutions

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$$S_{10} = \frac{10}{2}[2a + 9d] = 145$$

$$2a + 9d = 29 \dots (2)$$

From (1) and (2) $a = -8, d = 5$

$$S_{15} - S_5 = \frac{15}{2}[2a + 14d] - \frac{5}{2}[2a + 4d]$$

$$= \frac{15}{2}[-16 + 70] - \frac{5}{2}[-16 + 20]$$

$$= 405 - 10$$

$$= 395$$

Q13

$$\alpha = 1^2 + 4^2 + 8^2 \dots$$

$$t_n = a^2 + bn + c$$

$$1 = a + b + c$$

$$4 = 4a + 2b + c$$

$$8 = 9a + 3b + c$$

On solving we get, $a = \frac{1}{2}, b = \frac{3}{2}, c = -1$

$$\alpha = \sum_{n=1}^{10} \left(\frac{n^2}{2} + \frac{3n}{2} - 1 \right)^2$$

$$4\alpha = \sum_{n=1}^{10} (n^2 + 3n - 2)^2, \beta = \sum_{n=1}^{10} n^4$$

$$4\alpha - \beta = \sum_{n=1}^{10} (6n^3 + 5n^2 - 12n + 4) = 55(353) + 40$$

Q14

$$1^{\text{st}} \text{ GP} \Rightarrow t_1 = a, t_3 = b = ar^2 \Rightarrow r^2 = \frac{b}{a}$$

$$t_{11} = ar^{10} = a(r^2)^5 = a \cdot \left(\frac{b}{a} \right)^5$$

$$2^{\text{nd}} \text{ G.P.} \Rightarrow T_1 = a, T_5 = ar^4 = b$$

$$\Rightarrow r^4 = \left(\frac{b}{a} \right) \Rightarrow r = \left(\frac{b}{a} \right)^{1/4}$$

$$T_p = ar^{p-1} = a \left(\frac{b}{a} \right)^{\frac{p-1}{4}}$$

$$t_{11} = T_p \Rightarrow a \left(\frac{b}{a} \right)^5 = a \left(\frac{b}{a} \right)^{\frac{p-1}{4}}$$

$$\Rightarrow 5 = \frac{p-1}{4} \Rightarrow p = 21$$

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Solutions

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Q15

$$S_n = 3 + 7 + 11 + \dots n \text{ terms}$$

$$= \frac{n}{2}(6 + (n-1)4) = 3n + 2n^2 - 2n$$

$$= 2n^2 + n$$

$$\sum_{k=1}^n S_k = 2 \sum_{k=1}^n K^2 + \sum_{k=1}^n K$$

$$= 2 \cdot \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}$$

$$= n(n+1) \left[\frac{2n+1}{3} + \frac{1}{2} \right]$$

$$= \frac{n(n+1)(4n+5)}{6}$$

$$\Rightarrow 40 < \frac{6}{n(n+1)} \sum_{k=1}^n S_k < 42$$

$$40 < 4n + 5 < 42$$

$$35 < 4n < 37$$

$$n = 9$$

Q16

General term of the sequence,

$$T_r = \frac{r}{1 - 3r^2 + r^4}$$

$$T_r = \frac{r}{r^4 - 2r^2 + 1 - r^2}$$

$$T_r = \frac{r}{(r^2 - 1)^2 - r^2}$$

$$T_r = \frac{r}{(r^2 - r - 1)(r^2 + r - 1)}$$

$$T_r = \frac{\frac{1}{2}[(r^2 + r - 1) - (r^2 - r - 1)]}{(r^2 - r - 1)(r^2 + r - 1)}$$

$$= \frac{1}{2} \left[\frac{1}{r^2 - r - 1} - \frac{1}{r^2 + r - 1} \right]$$

Sum of 10 terms,

$$\sum_{r=1}^{10} T_r = \frac{1}{2} \left[\frac{1}{-1} - \frac{1}{109} \right] = \frac{-55}{109}$$

Q17

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Solutions

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 $1 + d, 1 + 7d, 1 + 43d$ are in GP

$$(1 + 7d)^2 = (1 + d)(1 + 43d)$$

$$1 + 49d^2 + 14d = 1 + 44d + 43d^2$$

$$6d^2 - 30d = 0$$

$$d = 5$$

$$S_{20} = \frac{20}{2} [2 \times 1 + (20 - 1) \times 5]$$

$$= 10[2 + 95]$$

$$= 970$$

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